

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018 – 2019 (Even Semester)

Innovativeness in Faculty Learning & Teaching and Learning outcome by Students

Degree, Semester & Branch: IV Semester B.E. ECE A

Course Code & Title: EC8453 Linear Integrated Circuits

Name of the Faculty member: Dr.S.Periyanayagi

Innovative Method	Topic	Effectiveness
Visualization Introduction – PPT	Active filters, Application of PLL	To provide Clear idea of the concept
Videos	Fabrication of a chip from sand	Gives an idea about manufacturing of ICs
Analogy	Analog filters, Comparators	Explaining the concept of Analog filter (new information) with water filter (Known information) helps them to capture the concept quickly.
Application oriented teaching	Introduction to Integrated Circuits, Operational Amplifiers	Explaining the concepts with real time applications creates interest among the students.
Simulation tool (Pspice)	Applications of OP-AMP, ADC and DAC	The students are made to simulate ADC and DAC using Pspice and analyze the waveform helped them to get a clear idea about the concept.
NPTEL Video	Operation of the basic PLL	Students are exposed to different mode of teaching.

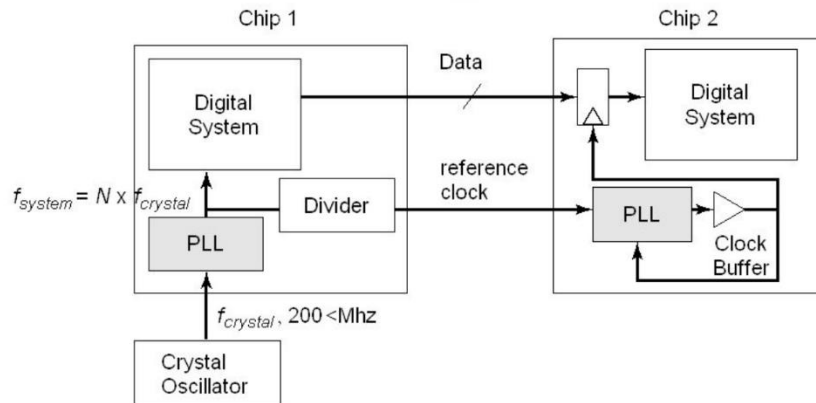
Using the concepts of online course in the subject:

Online Course: MITx: 3.15.2x Optical Materials and Devices

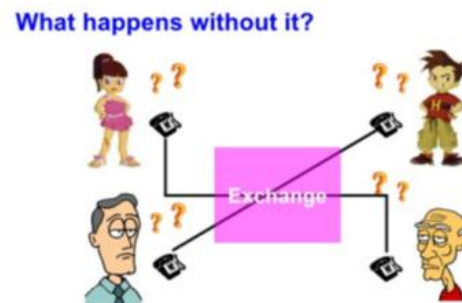
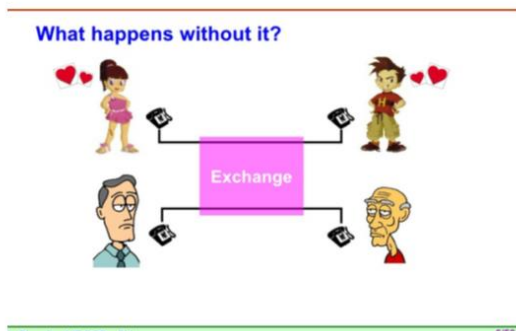
Course Code and Title	Lecture No.	Title
EC6404 Linear Integrated Circuits	58	Opto Couplers IC
	60	Fiber optic IC

Application of PLL :

PLL-Based Clock Synchronization



Synchronization



Importance of synchronization

References:

1. D.Roy Choudhry, Shail Jain, —Linear Integrated Circuits, New Age International Pvt. Ltd., 2018, Fifth Edition.
2. S.Salivahanan & V.S. Kanchana Bhaskaran, —Linear Integrated Circuits, TMH, 2nd Edition, 4th Reprint, 2016.
3. www.slideshare.com
4. <https://www.youtube.com/watch?v=bJNDh46uI3w>
5. <https://www.youtube.com/watch?v=Q5paWn7bFg4>